

THIAGO SAUD

Senior AI & MLOps Engineer — Principal Front-End Engineer

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PROFESSIONAL SUMMARY

Senior Software Engineer (7+ years) and Expert in high-performance, mission-critical front-end architecture with a proven track record of direct financial impact in ecosystems moving trillions of reals with millions of active clients (BTG Pactual & XP Investimentos). Direct reporting to C-Level and active technical mentorship of engineers. Actively transitioning to AI & MLOps Engineer, combining technical rigor in developing Machine Learning, Deep Learning, GenAI, LangChain, LangGraph pipelines and RAG architectures.

CORE COMPETENCIES

- **AI, MLOps, GenAI, LLMs & Computer Vision:** Machine Learning · Deep Learning · Computer Vision · Facial Recognition · AR/VR · PyTorch · TensorFlow · Scikit-learn · Hugging Face · MLflow · AWS SageMaker · Azure ML · Model CI/CD · LangGraph · LangChain · LlamaIndex · RAG · Fine-tuning (LoRA/QLoRA) · ChromaDB · FAISS · ONNX · NLU/NLP · MCP · Autonomous Agents · Prompt & Harness Engineering.
- **Architecture & Design Patterns:** Domain-Driven Design (DDD) · Test-Driven Development (TDD) · Clean Architecture · SOLID · Design System · System Design · Micro-frontends · Clean Code.
- **Data Engineering & Backend:** FastAPI · SQL · PostgreSQL · REST · GraphQL · WebSockets · UV · TOML.
- **DevOps, SecOps & Quality:** Docker · AWS (S3, CloudFront) · GitHub Actions · Azure DevOps · SonarQube · Datadog · Sentry · Core Web Vitals · Playwright · Vitest · Jest · Pytest · OAuth 2.0/JWT · OWASP Top 10.
- **Stacks & Leadership:** Python · TypeScript · JavaScript (ES6+) · Java · C/C++ · Angular · React · Vue.js · Next.js · Vite · Storybook · RxJS · NgRx · Mentorship · Code Review · ADR & RFC · Scrum/Kanban.

PROFESSIONAL EXPERIENCE

BANCO BTG PACTUAL S.A. — São Paulo, SP, BR (REMOTE)

Principal Front-End Engineer · Digital Equities Vertical | Feb 2023 – Jun 2026 (3 years and 5 months)

- **Promoted** in 5 months (2 years ahead of schedule) due to strategic impact, technical leadership and ownership.
- **Led** the architecture of mission-critical white-label financial platforms (**Home Broker Web & COE**) for high-frequency operations on B3, supporting 4.8 million customers, R\$ 1.6 trillion in AUM, 37% YoY high revenue and a 28.1% ROAE (Q3 2025).
- **Designed** component governance (Nx/Turborepo, Nexus, Storybook & Angular) and evolved the corporate Design System (ORQUESTRA), increasing code reuse by 60% and reducing visual defects by 35% (WCAG/A11Y).
- **Spearheaded** advanced rendering (GPU, Virtualization & Paging) and reactivity optimizations (Angular Signals & RxJS), cutting page load times by 45% and ensuring 60 FPS fluidity under real-time high-volatility data.
- **Adopted AI Engineering** across the development, DevOps, and SecOps lifecycle, reducing delivery time by 65% and code defects by 80% via LLMs, MCP, Cursor/Copilot, Autonomous Agents & Prompt & Harness Engineering.
- **Deployed** advanced micro-frontends architecture (Webpack Module Federation) backed by top-tier observability and security (Datadog, Sentry & OWASP Top 10).

XP INVESTIMENTOS S.A. — São Paulo, SP, BR (REMOTE)

Senior Front-End Engineer · Global Markets Vertical · Treasury | Jul 2021 – Nov 2022 (1 year and 5 months)

- **Architected** critical corporate financial platforms (**Black Sales & Black AAI**), supporting 12,000 advisors and 4.5 million active customers, R\$ 1.2 trillion in assets and R\$ 19.8 billion in consolidated gross revenue (Q4 2023).
- **Defended** complex architectural guidelines (RFCs & ADRs) directly to the Engineering Manager and C-Level (CFO/Treasury), standardizing 100% of legacy transition workflows and accelerating technical approvals by 40%.
- **Implemented** a scalable micro-frontends architecture (Webpack Module Federation), high-performance reactive flows (React, GraphQL, Redux-Saga, DDD & Clean Architecture) and advanced rendering optimizations (GPU & Virtualization & Pagination).
- **Evolved** the corporate Design System (SOMA), boosting component reuse by 50% and mitigating visual bugs by 33% (WCAG/A11Y).

ZOOX TECNOLOGIA LTDA. — Rio de Janeiro, RJ, BR (REMOTE)

Mid-Level Front-End Engineer | Jul 2020 – Nov 2020 (4 months)

- **Acted as a technical reference** in hospitality PMS white-label platform with direct stakeholder management and strategic alignment.
- **Modeled** critical facial recognition flows (Face matching & Bounding boxes) via AWS Rekognition, **achieving >90% accuracy and 100% regulatory compliance with LGPD and FNRH (Ministry of Tourism)**.
- **Structured** real-time billing architecture integrating global OTAs (Booking.com, Expedia & Agoda) via PostgreSQL, **eliminating overbooking operational failures and executing automated check-in/out transactions for thousands of daily rooms with Vuex**.

TG4 TECNOLOGIA LTDA. — Rio de Janeiro, RJ, BR (ON-SITE)

Junior Front-End Engineer | Sep 2019 – Nov 2020 (1 year and 2 months)

- **Partnered** directly with the CTIO developing high-volatility white-label chatbots for **5 major telecom operators (Tim, Claro, Vivo, Oi & Nextel)**.
- **Integrated NLU models** and predictive recommendation algorithms (Next Best Action), **doubling upgrade offer acceptance rates by 75% and driving key KPIs (CAC, CPC & LTV)** via (DDD, Clean Architecture, Angular, Docker & Playwright).

NEORIS DO BRASIL LTDA. — Rio de Janeiro, RJ, BR (HYBRID)

R&D Technology Intern | Apr 2017 – Apr 2019 (2 years)

- **Acted as a sole intern** reporting directly to 3 Senior Managers across **2 strategic large-scale products** for the client Aliansce (Shopping Center Administrator) and **2 complex internal PoCs**.
- **Developed** the "Aliansce BI Geospatial" web platform (Vanilla Javascript & Google Maps API) and **2 native Android AR apps** (Java & Vuforia SDK), alongside computer vision prototypes (Microsoft Azure Cognitive Services).

AI & MACHINE LEARNING PERSONAL PROJECTS

- [**CAUSAL TRANSFORMER MODEL \(SHAKESPEARE\)**](#): Pipeline in TensorFlow and Keras for stylized text generation (Shakespearean literary tone) with custom tokenization, checkpointing (TensorBoard), sampling (Temperature/Top-k) and Gradio. Built with dependency injection, strict typing and 100% test coverage via pytest.
- [**TECHNICAL REGRESSION PIPELINE \(BYD\)**](#): Advanced statistical predictive modeling (linear, multiple, and logistic regression) with scikit-learn and pandas applied to automotive specs. Automated generation of visual analytical reports and validation metrics (R^2 & Precision) tested with pytest.
- [**AUTONOMOUS AGENTS ECOSYSTEM**](#): Proprietary multi-layer Python agent architecture featuring Meteorological Agent (OpenWeather API with persistent memory), Restaurant Agent (PDFs & TF-IDF cosine similarity) and Intelligent Router. 100% test coverage via pytest.

EDUCATION & CERTIFICATIONS OF PROFESSIONAL SPECIALIZATIONS

- **BACHELOR OF COMPUTER SCIENCE — Rio de Janeiro, BR (ON-SITE)**
Universidade Veiga de Almeida (UVA) | Jan 2014 – Feb 2019 (5 years and 2 months)
Capstone: "Facial Recognition for Public Safety" (Python, OpenCV, Viola-Jones & PCA/LDA; Accuracy ~80–100%, response time ~0.8s).
- **BUSINESS ADMINISTRATION AND MANAGEMENT — Spain, Catalonia (REMOTE)**
Universitat de Barcelona (UB) | Jan 2018 – Nov 2019 (1 year and 11 months)
- **CERTIFICATIONS:** [GOOGLE: Foundations of User Experience Design \(2024\)](#) · [UNIVERSITY OF MICHIGAN: Python \(2026\)](#) · [ANTHROPIC: AI Fluency: Framework & Foundations \(2026\)](#) · [DEEPLARNING.AI: Deep Learning \(2026\)](#) · [IBM: AI Engineering \(2026\)](#) · [IBM: RAG & Agentic AI \(In progress\)](#) · [DUKE UNIVERSITY: MLOps | Machine Learning Operations \(In progress\)](#).
- **LANGUAGES:** Portuguese (Native) · English & Spanish (Limited working proficiency).